Compass Analytics
Know Your Customer

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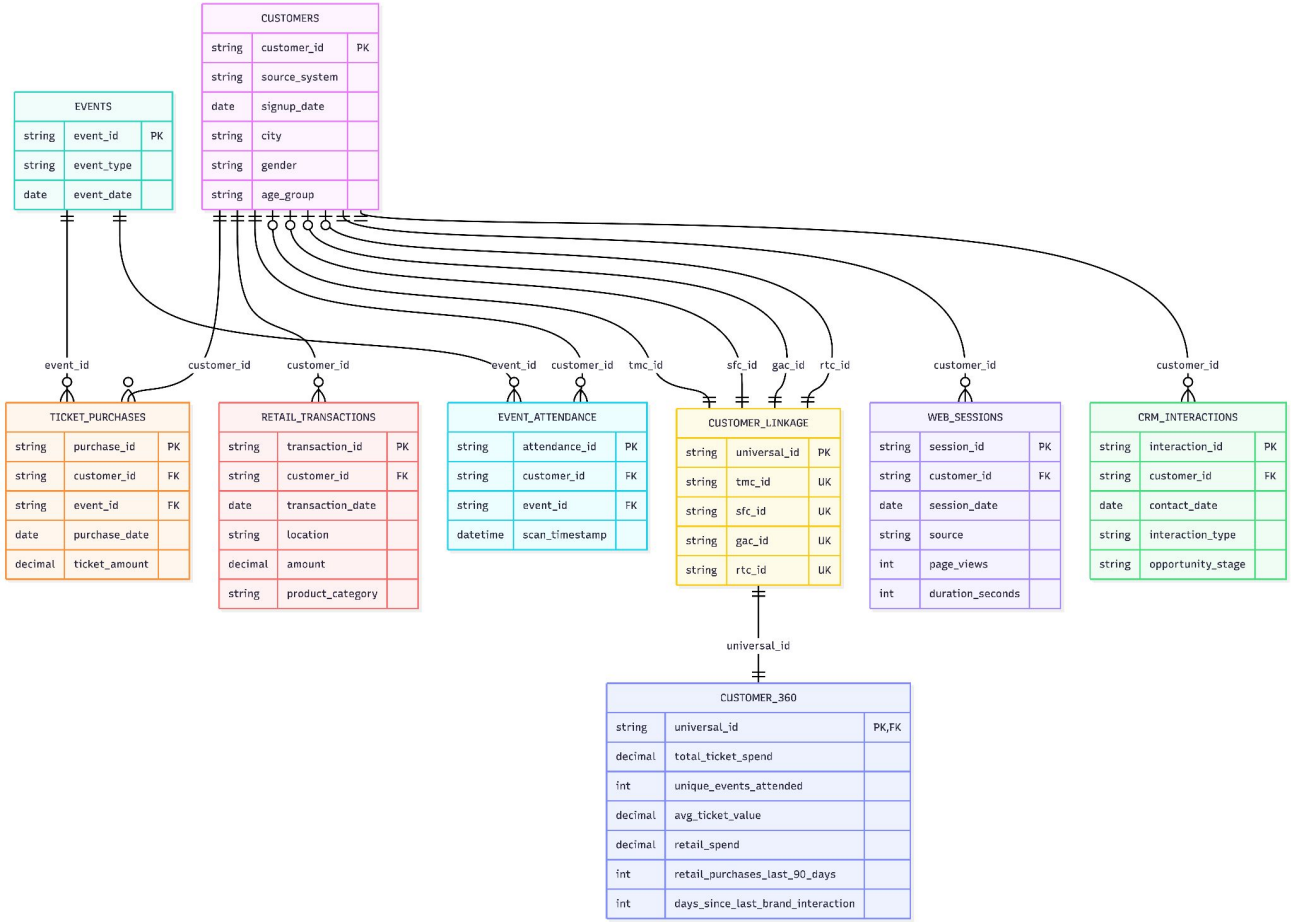
Task #1 – Identify Resolution Strategy (ERD)



** Map IDs by source system*

Task #1 – Identify Resolution Strategy (Pipeline)

1. Source Ingestion

Ingest + Standardize

- Load Ticketmaster, CRM, Web, and Retail data
- Standardize IDs, timestamps, and profile fields
- De-duplicate within each source system
- Identify unique source IDs (~40–55% lack matches)

Output → Standardized Source Records

2. Matching Sources

Match Across Systems

- Apply deterministic rules on strong identifiers
- Use probabilistic matching for unresolved records
- Score candidate matches by confidence
- Validate existing linkage and mismatches

Output → Candidate Customer Links

3. Audit Results

Audit + Validate

- Track match logic, confidence, and evidence
- Flag low-confidence or conflicting links
- Monitor quality and cluster consistency
- Create an auditable linkage layer

Output → Validate Identity Links

4. Resolve Gold Build

Build Unified Output

- Assign synthetic IDs to unresolved records
- Assign a `universal_id` to resolved customers
- Build the `customer_360` gold table
- Generate downstream analytical features

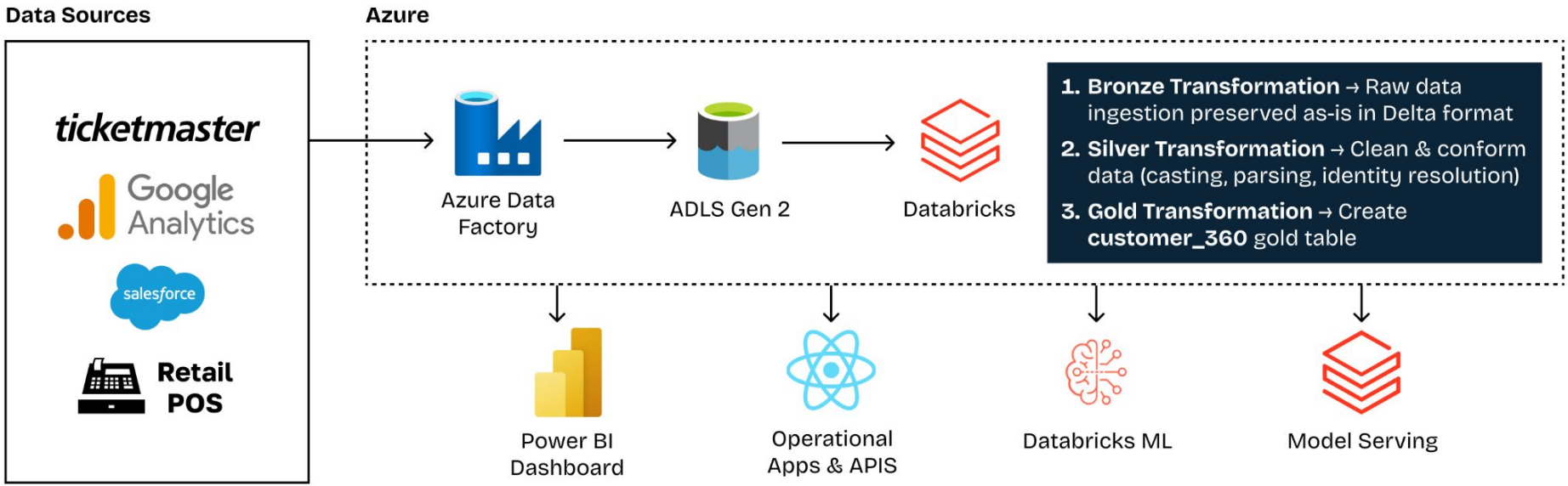
Output → Customer 360 Gold Table

Task #2 – Build Customer 360 Gold Table

1. **Unified Customer IDs**
   - Converted source-specific IDs into universal_id using customer_linkage
   - Preserved unmatched records as single-system customers
2. **Standardized Dates and Recency Logic**
   - Parsed all timestamps before feature engineering
   - Used the dataset max date as the recency reference point
3. **Built Required Feature Aggregates**
   - **Ticketing**: Spend, avg ticket value, purchase counts
   - **Attendance / Retail / Web / CRM**: Events attended, retail spend, etc.
4. **Built the Gold Table**
   - Outer-joined all channel features by `universal_id`
   - Added profile fields and days since last brand interaction
5. **Validated and Exported Output**
   - Checked totals, uniqueness, null handling, and feature sanity
   - Exported the final `customer_360` table for downstream use

Link to Notebook

Task #3 – Data Platform Overview (Architecture)



Key Points

1. Replace **cron jobs on VMs** with Azure Data Factory scheduled & event-driven triggers

2. Data governance with **Microsoft Purview and Unity Catalog** for role-based access

3. Databricks is **cloud-native, scalable, and can do both ML and AI,** and ADF connects to each source

Task #3 – Data Platform Overview (Detailed Overview)

Platform Choice

Azure Data Factory → Managed ETL service, replaces fragile cron-based pipelines

ADLS Gen 2 → Cost-effective object storage with ACID transactions via Delta format, enabling refinement

Databricks Lakehouse → Unified platform for data engineering and ML with great scalability

Databricks Lakehouse → Ensures lakehouse governance, replacing tribal knowledge with enforcable policy

Data Governance

Access Control → Role-based access via Unity Catalog and Microsoft Entra ID so teams only see relevant data

Microsoft Purview → Scans ADLS & Databricks assets for data governance issues

PIPEDA Compliance → Canadian privacy law requires consent tracking and right-to-deletion

Data Quality and Monitoring → Automated alerts on data refreshes, volume anomalies, and pipeline failures

Appendix